```
create database sales_management;

use sales_management;

-- 1 productline
create table productline(
productline_id int auto_increment primary key,
productline_name varchar (50) not null,
description varchar(255)
);
insert into productlines(productLine_name, description)
values
('Electronics', 'Electronic products'),
('Stationery', 'Office stationery items'),
('Bags', 'Different types of bags'),
('Bottles', 'Water bottles'),
('Accessories', 'Daily use accessories'),
('Furniture', 'Office furniture'),
('Kitchen', 'Kitchen products'),
('Sports', 'Sports products');

-- 2 products
create table products(
   product_id int auto_increment primary key,
   product_name varchar(50) not null,
   price decimal(10,2) not null,
   quantity int,
   productline_id int,
   foreign key (productline_id) references productline(productline_id)
   );
insert into products(product_name,price,productline_id,quantity)
values
```

```
('Laptop', 1, 45000.00, 10),
('Pen', 2, 20.00, 100),
('Notebook', 2, 60.00, 80),
('School Bag', 3, 800.00, 25),
('Water Bottle', 4, 250.00, 40),
('Headphones', 1, 1200.00, 30),
('Office Chair', 6, 3500.00, 15),
('Cricket Bat', 8, 1800.00, 12);

-- 3 offices
create table offices (
    office_id int auto_increment primary key,
    city varchar(50) not null,
    country varchar(50) not null,
    phone varchar(20),
    address varchar(150)
);
insert into offices (city, country, phone, address)
values
('Chennai', 'India', '9876543210', 'Anna Nagar'),
('Madurai', 'India', '9876543211', 'KK Nagar'),
('Coimbatore', 'India', '9876543212', 'Gandhipuram'),
('Bangalore', 'India', '9876543213', 'Indiranagar'),
('Hyderabad', 'India', '9876543214', 'Banjara Hills'),
('Mumbai', 'India', '9876543215', 'Andheri'),
('Delhi', 'India', '9876543216', 'Karol Bagh'),
('Pune', 'India', '9876543217', 'Shivaji Nagar');

-- 4 employees
create table employees (
    employee_id int auto_increment primary key,
    first_name varchar(50) not null,
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```
    last_name varchar(50) not null,
    job_title varchar(50),
    office_id int,
foreign key(office_id) references offices(office_id)
);
insert into employees (first_name, last_name, job_title, office_id)
values
('Arun', 'Kumar', 'Sales Manager', 1),
('Priya', 'Raj', 'Sales Executive', 1),
('Vijay', 'Anand', 'Sales Executive', 2),
('Divya', 'Sri', 'Accountant', 3),
('Karthik', 'Ravi', 'Sales Executive', 4),
('Meena', 'Devi', 'HR Manager', 5),
('Rahul', 'Sharma', 'Sales Manager', 6),
('Anitha', 'K', 'Sales Executive', 7);

-- 5 customers
create table customers (
    customer_id int auto_increment primary key,
    customer_name varchar(100) not null,
    phone varchar(20),
    city varchar(50),
    country varchar(50)
    );
    insert into customers (customer_name, phone, city, country)
    values
('Rahul', '9000000001', 'Chennai', 'India'),
('Priya', '9000000002', 'Madurai', 'India'),
('Arun', '9000000003', 'Coimbatore', 'India'),
('Divya', '9000000004', 'Chennai', 'India'),
('Karthik', '9000000005', 'Bangalore', 'India'),
('Meena', '9000000006', 'Hyderabad', 'India'),
```

```
('Vijay', '9000000007', 'Mumbai', 'India'),
('Anitha', '9000000008', 'Delhi', 'India');

-- 6 orders
 create table orders (
   order_id int auto_increment primary key,
   customer_id int not null,
   order_date date,
   status varchar(30),
   foreign key(customer_id) references customers(customer_id));

insert into orders (customer_id, order_date, status)
value
(1, '2026-09-01', 'Delivered'),
(2, '2026-09-02', 'Pending'),
(3, '2026-09-03', 'Delivered'),
(4, '2026-09-04', 'Shipped'),
(5, '2026-09-05', 'Delivered'),
(6, '2026-09-06', 'Pending'),
(7, '2026-09-07', 'Shipped'),
(8, '2026-09-08', 'Delivered');

-- 7 order details
create table orderdetails (
   orderdetail_id int auto_increment primary key,
   order_id int not null,
   product_id int not null,
   quantity int not null,
   priceEach decimal(10,2) not null,
   foreign key (order_id) references orders(order_id),
   foreign key (product_id) references products(product_id));
```

```
insert into orderdetails (order_id, product_id, quantity, priceEach)
values
(1, 1, 1, 45000.00),
(2, 2, 5, 20.00),
(3, 3, 3, 60.00),
(4, 4, 1, 800.00),
(5, 5, 2, 250.00),
(6, 6, 1, 1200.00),
(7, 7, 2, 3500.00),
(8, 8, 1, 1800.00);

-- 8 payments
create table payments (
   payment_id int auto_increment primary key,
   customer_id int not null,
   payment_date date,
   amount decimal(10,2) not null,
   payment_method varchar(30),
   foreign key (customer_id) references customers(customer_id)
);

insert into payments (customer_id, payment_date, amount, payment_method)
values
(1, '2026-09-01', 45000.00, 'UPI'),
(2, '2026-09-02', 100.00, 'Cash'),
(3, '2026-09-03', 180.00, 'Card'),
(4, '2026-09-04', 800.00, 'UPI'),
(5, '2026-09-05', 500.00, 'Cash'),
(6, '2026-09-06', 1200.00, 'Card'),
(7, '2026-09-07', 7000.00, 'UPI'),
(8, '2026-09-08', 1800.00, 'Card');

-- select all
```

```
select * from payments;

-- using where ,and
select payment_date, amount
from payments
where amount > 500 and customer_id > 2;

-- using orderby
select payment_date, amount
from payments
where amount > 500 and customer_id > 2
order by payment_date desc;

-- using limit
select payment_date, amount
from payments
where amount > 500 and customer_id > 2
order by payment_date desc
limit 3;

-- using alter
alter table payments
rename to payment;
alter table employees
ADD COLUMN reports_to int null,
add constraint fk_reports_to foreign key (reports_to) references employees(employee_id);

-- using aggregate functions
select count(*) as total
from payment;
select sum(amount) as total
from payment;
```

```
select avg(amount) as total
from payment;
select min(amount) as total
from payment;
select max(amount) as total
from payment;

-- aggregate functions with groupby
select customer_id, count(*)as custom
from payment
group by customer_id;
select order_id, sum(product_id) as custom
from orderdetails
group by order_id;
select order_id, min(product_id) as custom
from orderdetails
group by order_id;
select order_id, max(product_id) as custom
from orderdetails
group by order_id;

-- aggregate functions with having
select order_id, min(product_id) as custom
from orderdetails
group by order_id
having(custom)>2;
select order_id, max(product_id) as custom
from orderdetails
group by order_id
having(custom)>2;
select order_id, count(*) as custom
from orderdetails
```

```
group by order_id
having(custom)>2;
select order_id, sum(product_id) as custom
from orderdetails
group by order_id
having(custom)>2;

-- subquery
select product_name,price
from products
where price > (
select avg(price)
from products
);

-- joins

-- using inner join
select
customers.customer_id,
orders.order_id
from customers
inner join orders
on customers.customer_id=orders.customer_id;

-- left join
select
customers.customer_id,
orders.order_id
from customers
left join orders
on customers.customer_id=orders.customer_id;
```

```
-- right join
select
customers.customer_id,
orders.order_id
from customers
right join orders
on customers.customer_id=orders.customer_id;

-- full join
select
customers.customer_id,
orders.order_id
from customers
left join orders
on customers.customer_id=orders.customer_id
union
select
customers.customer_id,
orders.order_id
from customers
right join orders
on customers.customer_id=orders.customer_id;

-- cross join
select
customers.customer_id,
orders.order_id
from customers
cross join orders;

-- self join
```

```
select
e.first_name as employees,
m.first_name as manager
from employees e
left join employees m
on e.reports_to = m.employee_id;
```